**Project Design Phase-II**

**Technology Stack (Architecture & Stack)**

**Date:** 03 October 2026
**Team ID:** 6aba117e538da4003d63184c
**Project Name:** Script-Controlled ACL – Restrict Record Access Based on Field Value
**Maximum Marks:** 4 Marks

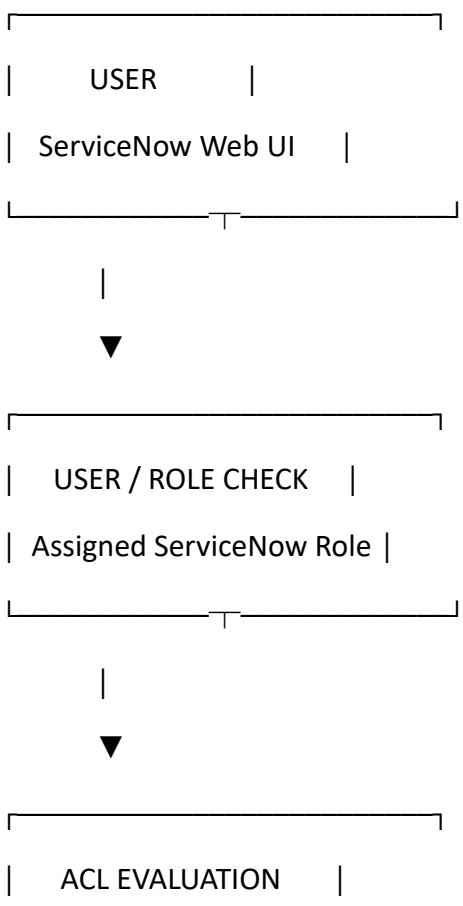
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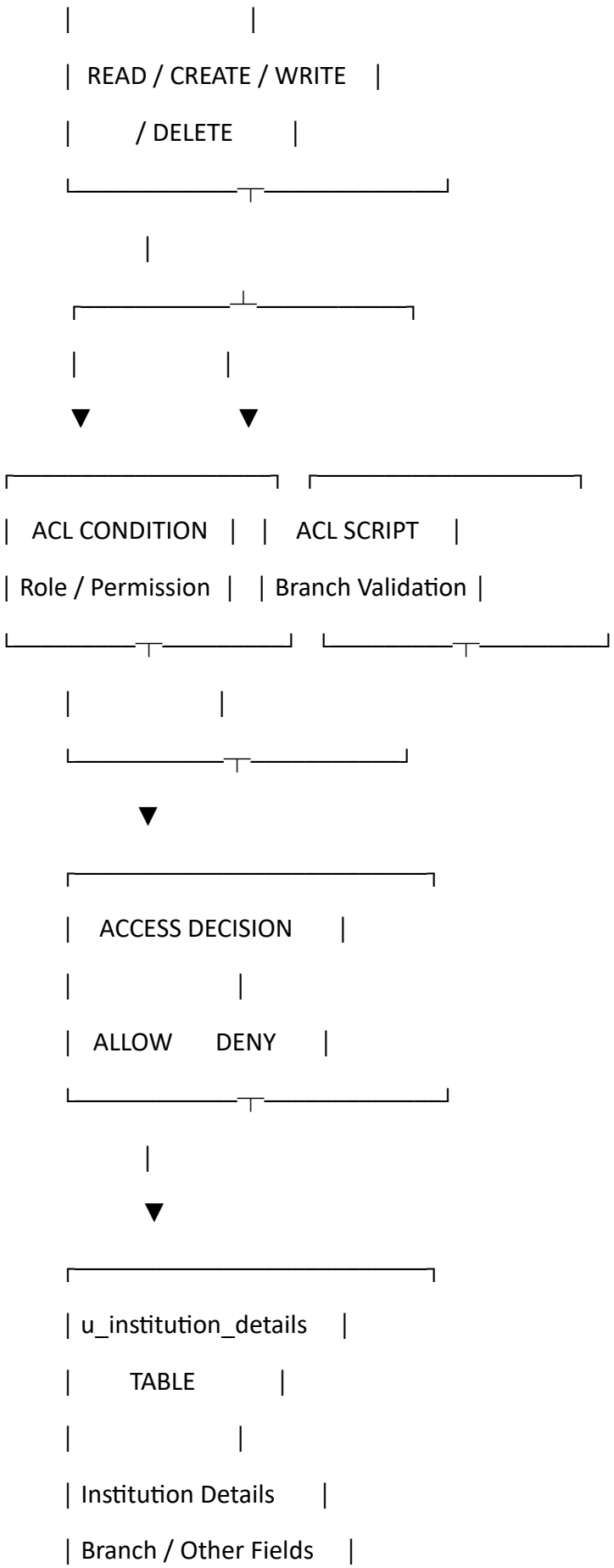
**Technical Architecture**

The proposed solution is implemented using the **ServiceNow platform**. Users access the u_institution_details table through the ServiceNow web interface. Based on the user's assigned role and the configured **Access Control Lists (ACLs)**, ServiceNow determines whether the user is permitted to perform operations such as **Read, Create, Write, and Delete**.

The ACL scripts additionally validate the **Branch** field to enforce field-based access restrictions. Authorized requests are allowed to proceed, while unauthorized requests are denied by the ACL mechanism.

**Architecture Flow**




READ / CREATE / WRITE / DELETE
ACL CONDITION
Role / Permission
ACL SCRIPT
Branch Validation
ACCESS DECISION
ALLOW
DENY
u_institution_details
TABLE
Institution Details
Branch / Other Fields

**Architecture Description**

1. **User Interface:** Users interact with the ServiceNow platform through its web interface.
2. **User and Role Management:** ServiceNow identifies the logged-in user and checks the roles assigned to the user.
3. **ACL Evaluation:** ServiceNow evaluates the applicable ACL rules for the requested operation.
4. **ACL Conditions:** Role-based conditions determine whether the user has the required permission.
5. **ACL Script:** A server-side script performs additional validation based on the record's **Branch** value.
6. **Access Decision:** The request is either allowed or denied.
7. **Database/Table:** Authorized users can access or modify records in the u_institution_details table according to their permissions.

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**Table-1: Components & Technologies**

| S.No | Component | Description | Technology |
|---|---|---|---|
| **1** | **User Interface** | Interface through which users access and manage institution detail records. | ServiceNow Web UI |
| **2** | **Application Logic-1** | User and role-based access management for controlling permissions. | ServiceNow Users & Roles |
| **3** | **Application Logic-2** | ACL evaluation for Read, Create, Write and Delete operations. | ServiceNow ACL |
| **4** | **Application Logic-3** | Script-based validation of the Branch field to restrict unauthorized record access. | ServiceNow Server-side JavaScript |
| **5** | **Database** | Stores institution details, user information, roles and configuration data. | ServiceNow Platform Database |
| **6** | **Cloud Database** | Not separately configured; database services are managed internally by the ServiceNow platform. | ServiceNow Managed Database |

| S.No | Component | Description | Technology |
|---|---|---|---|
| 7 | **File Storage** | No dedicated file-storage component is required for the ACL solution. | Not Applicable |
| 8 | **External API-1** | No external API is required for implementing the access-control mechanism. | Not Applicable |
| 9 | **External API-2** | No external third-party API is used in the proposed solution. | Not Applicable |
| 10 | **Machine Learning Model** | Machine learning is not required for the proposed ACL-based access-control solution. | Not Applicable |
| 11 | **Infrastructure (Server / Cloud)** | The application and ACL configuration execute within the ServiceNow cloud platform. | ServiceNow Cloud Platform |

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**Table-2: Application Characteristics**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | **Open-Source Frameworks** | No separate open-source application framework is required. The solution uses the ServiceNow platform and its native development capabilities. | ServiceNow Platform / JavaScript |
| 2 | **Security Implementations** | Access is controlled using users, roles, ACLs, conditions and server-side scripts. Unauthorized users are prevented from performing restricted operations. | ServiceNow ACL, Roles, Conditions, Server-side JavaScript |
| 3 | **Scalable Architecture** | ACL rules can be extended to additional users, roles, tables, branches and operations without redesigning the complete application. | ServiceNow Platform, ACL Framework |
| 4 | **Availability** | The solution operates within the ServiceNow cloud environment, allowing users to access the application through the ServiceNow platform whenever the platform is available. | ServiceNow Cloud Platform |

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 5 | **Performance** | ACL conditions and scripts are designed to perform access validation during record operations with minimal processing overhead. Unnecessary external services and complex processing are avoided. | ServiceNow ACL Engine, Server-side JavaScript |

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**Technology Stack Summary**

| Layer | Technology |
|---|---|
| **Platform** | ServiceNow |
| **User Interface** | ServiceNow Web UI |
| **Access Control** | ServiceNow ACL |
| **Authentication** | ServiceNow User Authentication |
| **Authorization** | ServiceNow Roles + ACL |
| **Business Logic** | ServiceNow Server-side JavaScript |
| **Data Table** | u_institution_details |
| **Database** | ServiceNow Managed Database |
| **Testing** | ServiceNow Interface / ACL Testing |
| **External APIs** | None |
| **Machine Learning** | None |
| **Deployment** | ServiceNow Cloud Platform |